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隱形帳簿：量化東協未計量的平台經濟

The Invisible Ledger: Quantifying ASEAN's Unmeasured Platform Economy

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Abstract

This proposal examines an administrative information gap in Southeast Asia's platform economy: the difference between gross transaction value processed through digital platforms and the revenue booked by the platforms themselves. The study calls this difference the "invisible wedge" and measures its scale with an ecosystem ratio, defined as unbooked gross transaction value divided by platform revenue. Substantial empirical work has already been completed. Using fiscal-year 2023 disclosures from Grab, GoTo, and Shopee, the current Indonesia-focused base estimate is $66.8 billion of unbooked GTV with an ecosystem ratio of 16.0×. The study also examines whether the transaction-revenue gap persists within platform disclosures, uses Malaysia's 2024 low-value-goods regime as a descriptive policy benchmark, calibrates an ASEAN-5 order-of-magnitude scenario, and tests whether ecosystem-ratio information is associated with short-window investor reactions around earnings announcements. It does not interpret unbooked GTV as profit, unpaid tax, or GDP value-added. The objective is to quantify a digitally recorded transaction flow that may remain outside routine participant-level third-party reporting and to assess its implications for fiscal capacity, platform reporting, and financial interpretation.

1. Introduction

Southeast Asia's digital economy grew from $98 billion in 2020 to $263 billion in 2024 (Google, Temasek and Bain 2024). Platform scale, however, is not the same as tax-administrative visibility. Platforms record transactions, determine participant payouts, and book their own revenue, while the gross receipts earned by merchants, drivers, and other participants may remain outside routine third-party income reporting. I call this administrative information gap the "invisible wedge." The empirical question is whether this digitally recorded but separately reported flow is large enough to matter for tax administration.

This study estimates that wedge for an Indonesia-focused sample of Grab, GoTo, and Shopee, then uses the resulting retained share to calibrate four additional Association of Southeast Asian Nations (ASEAN) markets: Vietnam, the Philippines, Thailand, and Malaysia.

The term invisible refers to the administration's information channel, not to criminal concealment. Suppose a platform processes 100 units of transaction value and books 10 as its own revenue. The remaining 90 is still recorded in the platform's system, but it is not platform revenue; it may include merchant receipts, driver payouts, inventory costs, taxes paid elsewhere, and other pass-through amounts. The study measures this accounting gap without assuming that the residual is profit, taxable income, or unpaid tax.

I measure the wedge with an ecosystem ratio: gross transaction value (GTV) not booked as platform revenue, divided by the revenue the platform does book. The core estimate uses fiscal-year 2023 disclosures. Grab and Shopee inputs are Indonesia-specific or transparently derived for Indonesia; GoTo reports the required inputs directly only at Group level, which also includes activity outside Indonesia. The $66.8 billion result is therefore an Indonesia-focused base estimate, while the four additional country figures are scenario calibrations rather than independent measurements.

Grab, GoTo, and Shopee are useful empirical settings because they directly intermediate transactions between consumers and merchants or service providers and publish financial disclosures from which transaction value and platform revenue can be observed or transparently derived. Their role as both transaction intermediary and record keeper places them directly at the center of the study's third-party-reporting question.

The study measures gross participant-facing transaction value that is not booked as platform revenue; it does not estimate participant profit, unpaid tax, or value-added.

Accordingly, the regional figure is an order-of-magnitude scenario anchored on the Indonesia-focused base, not five independently measured country wedges.

1.1 Research Questions

The proposal organizes the thesis around four research questions. The first two concern measurement and persistence, the third concerns policy relevance, and the fourth asks whether the measured ratio has short-window financial-market relevance.

Table 1. Research questions and empirical role

Research question	Empirical role
RQ1. How large is the transaction-revenue wedge in the 2023 Indonesia-focused Grab, GoTo, and Shopee sample?	Core measurement of the invisible wedge and ecosystem ratio.
RQ2. Does the gap persist within scope-matched platform disclosures over time?	Historical robustness using GoTo, Grab, and Shopee primary disclosures.
RQ3. What can Malaysia's 2024 low-value-goods expansion show descriptively about expanding digitally observable tax bases?	Limited policy benchmark; no causal treatment-effect claim.
RQ4. Is the ecosystem ratio associated with short-window market reactions around earnings announcements?	Finance test using market-adjusted event returns.

1.2 Expected Contributions and Claim Boundaries

The study is designed to make four contributions. First, it constructs a 2023 Indonesia-focused platform estimate and uses the combined 5.88% take rate only as a transparent calibration for four additional ASEAN markets. Derived and Group-level inputs are explicitly identified by provenance.

Second, I use Malaysia's January 2024 low-value-goods expansion as a descriptive policy benchmark, reporting both the policy's observed collections and a two-country before-and-after comparison with Vietnam without a causal claim. Third, I rebuild GoTo's 2020-2023 Group series from primary filings. The unbooked share remains between 97.6% and 98.9%, showing a persistent transaction-revenue gap rather than a monotonic increase.

Fourth, I use scope-matched historical Grab and Shopee disclosures to test whether large transaction-to-revenue gaps persist within platforms, and a 47-event earnings-announcement panel to test short-window investor response. The event study finds no stable association between the ecosystem ratio and market-adjusted cumulative abnormal returns.

These contributions are deliberately bounded. The regional total is a calibrated scenario rather than five independent country measurements; the Malaysia exercise is descriptive rather than causal; and the ecosystem ratio is an accounting and administrative-visibility measure rather than an estimate of taxable income, value-added, evasion, or tax due.

2. Literature Review

2.1 Platform economics and the location of value

The economics of multisided platforms begins with the interaction of distinct user groups whose participation affects the value received by the other side. Rochet and Tirole (2003), Caillaud and Jullien (2003), Parker and Van Alstyne (2005), and Armstrong (2006) show why platform pricing cannot be read as a simple markup over cost: prices and commissions are chosen to balance cross-group network effects, participation, and competition. Hagiu and Wright (2015) further distinguish a multisided platform from a vertically integrated firm by locating decision rights and entrepreneurial effort with affiliated participants rather than inside the platform. Evans and Schmalensee (2016) translate the same point into the business-model literature. These results explain why low retained shares can coexist with large ecosystems and why platform revenue is a poor proxy for the transaction value coordinated through the platform.

2.2 Informality, platform work, and measurement

Traditional informal-economy estimates rely on household surveys, firm surveys, or indirect macroeconomic indicators. Medina and Schneider (2019) estimate a large global shadow economy; La Porta and Shleifer (2014) emphasise the small scale and low productivity of informal firms; and Ulyssea (2018) shows that registration and off-the-books employment are distinct margins whose welfare effects cannot be inferred from formal status alone. The International Labour Organization (2021, 2023) likewise treats platform work and informality as overlapping but non-identical categories.

Platform participants occupy an unusual position in this literature. Their transactions are priced, recorded, and settled through a formal intermediary, but the participant may remain self-employed, unregistered, below filing thresholds, or simply outside an automatic reporting channel. The object measured here is therefore not the shadow economy as conventionally defined. It is a digitally recorded flow whose administrative treatment resembles self-reported income even though an intermediary already possesses the record.

2.3 Third-party reporting as tax capacity

A central result in public finance is that tax compliance depends strongly on third-party information and remittance structure. Kleven et al. (2011) find an evasion rate of 0.3

percent on income subject to third-party reporting versus 37 percent on self-reported income in Denmark. Pomeranz (2015) shows that the VAT paper trail creates self-enforcement across firms, while Naritomi (2019) demonstrates that consumer-held records can raise reported sales. Kleven, Kreiner and Saez (2016) formalise firms as fiscal intermediaries, and Slemrod (2019) places information reporting and remittance regimes among the core instruments of modern enforcement.

Jensen (2022) connects this mechanism to development: as employment moves into organisations that keep records and report income, the feasible tax base expands. Platform work creates a new configuration - a large intermediary with detailed records but, in many jurisdictions, no comprehensive obligation to report seller or worker income. Collins et al. (2019) and Garin et al. (2023) show how platform information returns affect the measured prevalence of gig work in the United States. The administrative problem is therefore not a lack of data generation; it is the absence, incompleteness, or fragmentation of the reporting rule.

Two finance studies extend this administrative-data channel. Barrios, Hochberg, and Yi (2022) show that gig-platform entry increases new business formation and small-business lending, while Denes, Lagaras, and Tsoutsoura (2025) use U.S. tax-return microdata to show that gig work can serve as a pathway into entrepreneurship. Together, these studies reinforce the idea that platform-mediated work leaves economically meaningful financial and administrative records.

2.4 Digital taxation and platform-reporting rules

Corporate digital-tax reforms and platform-participant reporting address different bases. The OECD Pillar One blueprint concerns the allocation of taxing rights over large multinational enterprises (OECD 2020b). By contrast, the OECD's Forum on Tax Administration report and Model Rules require platforms to collect and report information on income realised by sellers and service providers (OECD 2019, 2020a). The European Union's DAC7 regime similarly imposes due-diligence and reporting duties on platform operators (European Union 2021). These instruments provide the closest policy precedent for the mechanism studied here: using the platform's existing transaction record to improve participant-level visibility without treating platform revenue as the participant tax base.

3. Conceptual Framework: The Ecosystem Ratio

Return to the schematic from the Introduction: if a platform processes 100 units of transaction value and books 10 as revenue, 90 units pass through the ecosystem without being recorded as the platform's own revenue. The ecosystem ratio formalizes that comparison: (100 − 10) / 10 = 9.

Gross transaction value (GTV), or gross merchandise value (GMV) in some platform disclosures, is the total value of transactions processed through a platform, regardless of who ultimately receives the proceeds. This construction is consistent with the platform-economics literature showing that booked platform revenue need not track the total transaction value coordinated through a multisided platform (Rochet and Tirole 2003; Evans and Schmalensee 2016).

The ecosystem ratio describes the scale of participant-facing transaction value relative to the platform's auditable revenue base. A higher ratio indicates a larger unbooked transaction flow relative to the platform's own revenue; it does not by itself imply greater participant profit, taxable income, value-added, non-compliance, or tax due.

This distinction is essential. Platform filings and investor disclosures can make the accounting gap measurable, but the tax treatment of the residual depends on participant type, deductible costs, thresholds, registration status, and taxes already remitted through other channels. The empirical contribution is to quantify the size of the recorded flow that current corporate-level digital taxes do not directly reach.

Ecosystem ratio = (GTV - platform revenue) / platform revenue

4. Data and Methodology

4.1 Platform Selection and Institutional Background

I focus on Grab, GoTo, and Shopee (operated by Sea Limited) because they are large ASEAN-origin consumer platforms that directly intermediate transactions and provide public financial disclosures sufficient to observe or transparently derive transaction value and platform revenue. Public listing makes these platforms unusually observable relative to private competitors, which is essential for the company-level measurement used here.

Grab Holdings Limited. Grab began as a ride-hailing app in Malaysia in 2012 and now operates Mobility, Deliveries, and Financial Services across Southeast Asia. It listed on

Nasdaq in December 2021 through a special-purpose acquisition company (SPAC) merger. For this study, Grab's relevant transaction measure is On-Demand GMV, which combines Mobility and Deliveries; Financial Services is reported separately and is excluded from that GMV measure.

PT GoTo Gojek Tokopedia Tbk (GoTo). GoTo was formed through the 2021 merger of Gojek and Tokopedia and listed on the Indonesia Stock Exchange in April 2022. Indonesia is its core market, but Group disclosures are not a country-level split. From 2024, Tokopedia's e-commerce operations were deconsolidated after the TikTok Shop transaction, creating a structural break in GoTo's reported GTV and revenue series. Pre- and post-2024 figures are therefore not fully like-for-like.

Sea Limited / Shopee. Sea Limited, headquartered in Singapore, operates Shopee (e-commerce), Garena (digital entertainment), and Monee, formerly SeaMoney (digital financial services). Sea listed on the NYSE in 2017. Because consolidated Sea revenue includes all three businesses, the analysis pairs Shopee GMV with Shopee e-commerce revenue whenever the filings permit rather than mixing a segment numerator with a consolidated denominator.

Scope note. These platforms do not provide interchangeable measurements of one common business model. Grab and GoTo are centered on on-demand services, while Shopee is centered on e-commerce; their geographic coverage and revenue-recognition conventions also differ. For that reason, the 2023 base is described as Indonesia-focused, cross-platform ratio levels are interpreted cautiously, and the four non-Indonesian country figures are treated as calibrations rather than direct measurements.

4.2 Sample Construction and Data Provenance

Platform revenue and transaction value are taken from primary company disclosures: Grab and Sea Limited Form 20-F filings and investor releases, and GoTo annual and quarterly reports. Momentum Works (2024) is used only where an Indonesia-specific market-share input is not separately disclosed. The core cross-platform sample is fiscal year 2023, and Table 2 identifies whether each input is directly reported, derived for Indonesia, or reported only at GoTo Group level.

Take rate is platform revenue divided by GTV. The base estimate's blended take rate is the same ratio computed across all three platform rows combined—total platform revenue divided by total GTV, 5.88%—and is used only for the four-country regional scenario calibration. Because the GoTo row is Group-level, I refer to this as the Indonesia-focused blended take rate rather than a perfectly country-isolated Indonesia parameter.

A separate historical sample tests persistence within platforms without mixing geographic scopes. It extends scope-matched Grab Mobility + Deliveries data through 2019-2021 and Shopee GMV and segment revenue through 2017-2021; these observations are reported separately from the Indonesia-focused Table 3.

Table 2. Source, derivation, and geographic scope of the 2023 Indonesia-focused inputs

Platform	Metric	Source	Provenance
Grab	Revenue	Grab 2023 Form 20-F	Direct country-level disclosure.
Grab	GTV	Derived	Indonesia revenue divided by company-wide take rate (11.24%); country GTV is not independently disclosed.
GoTo Group	GTV and net revenue	GoTo FY2023 annual report	Both inputs directly reported at Group level; not country-isolated.
Shopee	GTV and revenue	Sea 2023 Form 20-F; Momentum Works 2024	Indonesia GMV estimated from market share; revenue derived from disclosed take rate (10.06%).

Only GoTo supplies both inputs independently, but its disclosures are Group-level. Grab and Shopee Indonesia-specific rows therefore inherit transparent derivations where country-level transaction values are not separately disclosed.

4.3 Core Measurement and Sensitivity

Table 3 establishes the 2023 Indonesia-focused base. GoTo's ecosystem ratio (40.0×) is much higher than Grab's and Shopee's (roughly 8-9×), largely because GoTo reports full marketplace GTV while booking only a thin commission as revenue. The cross-platform spread therefore reflects reporting convention as well as economic structure. The combined 5.88% take rate is used only for the regional scenario calibration.

Table 4 varies the Indonesia-focused blended take rate from 5% to 10% while holding total GTV fixed. It isolates sensitivity to the retained-share assumption; uncertainty in the platform-specific derivations for Grab and Shopee is not separately propagated.

Table 3. Indonesia-focused platform estimate, FY2023

Platform	GTV	Net revenue	Unbooked GTV	Ecosystem ratio
Grab	$5.40B	$0.61B	$4.80B	7.9×
GoTo Group	$39.81B	$0.97B	$38.84B	40.0×
Shopee	$25.80B	$2.60B	$23.20B	8.9×
Total	$71.00B	$4.18B	$66.83B	16.0×

Grab and Shopee rows are Indonesia-specific or transparently derived for Indonesia. GoTo uses directly reported Group GTV and net revenue because no country-level GTV split is disclosed. Ratios use unrounded inputs; displayed values are rounded.

Table 4. Sensitivity of the core estimate to the assumed blended take rate

Take rate	Platform revenue	Unbooked GTV	Ecosystem ratio
5.00%	$3.55B	$67.45B	19.0×
5.88% (base)	$4.18B	$66.83B	16.0×
7.50%	$5.33B	$65.68B	12.3×
10.00%	$7.10B	$63.90B	9.0×

The dollar wedge changes less than the ratio because the retained share is small; the ratio is largely a transformation of the retained-share assumption.

4.4 Historical Persistence

A separate within-platform exercise tests whether the transaction-revenue gap is persistent rather than a one-year artifact. GoTo's 2020-2023 Group series is rebuilt from primary disclosures, while Grab and Shopee historical observations are kept on scope-matched segment or company bases. These series are not treated as Indonesia-specific time series.

4.5 Malaysia Policy Benchmark and Regional Calibration

Malaysia expanded its digital tax base in January 2024 through the Low Value Goods (LVG) regime (Royal Malaysian Customs Department 2023; Ministry of Finance, Malaysia 2023). I use two descriptive benchmarks: the LVG regime's own reported collections and a before-and-after comparison with Vietnam. They answer different questions and are not combined into a single treatment-effect estimate.

The regional exercise separately applies the Indonesia-focused base estimate's 5.88% retained share to 2024 digital-economy GMV in Vietnam, the Philippines, Thailand, and Malaysia. These four rows are scenario calibrations rather than independent country-level measurements; their identical 16.0× ratios are arithmetic consequences of using one retained-share assumption.

4.6 Investor-Response Event Study

The investor-response analysis uses a 47-firm-quarter event panel: 17 Grab quarters, 16 GoTo quarters, and 14 Sea quarters from 2022Q1 through 2026Q1, subject to disclosure availability. Each observation uses contemporaneously disclosed, scope-matched GMV/GTV and revenue. Quarter-over-quarter change is calculated only when the immediately preceding calendar quarter is available, leaving 43 valid adjacent-quarter changes. Event returns are centered on the first trading session able to react to the release rather than mechanically on the publication date.

The event-study component asks a narrow question: when an earnings release discloses GMV/GTV and revenue, is the resulting ecosystem ratio associated with the platform's short-window stock-market reaction? Earnings announcements provide the natural event point because the ratio becomes observable when those inputs are released.

I use a market-adjusted event study (Brown and Warner 1985; MacKinlay 1997). Abnormal return is the platform's daily close-to-close return minus the matched market-index return: the NASDAQ Composite for Grab (Nasdaq: GRAB), the Jakarta Composite Index (JCI) for GoTo (IDX: GOTO), and the S&P 500 for Sea (NYSE: SE). CAR[-1,+1] is centered on the first trading session able to react to the disclosure; after-close releases are shifted to the next trading session. The ecosystem ratio is tested in levels and as true adjacent-quarter percentage change, with pooled levels interpreted cautiously because reporting scope differs across platforms.

Under the strict primary rule, a quarter is contaminated only when the same earnings release revises previously issued guidance or announces a genuinely new major corporate event, such as M&A, a capital raise, a new repurchase program, or a major restructuring. Initial guidance is coded as a candidate: it remains in the strict primary sample and is removed only in the broader robustness check. Updates on previously announced transactions or repurchase programs do not count as new contamination events.

The event panel is small and unbalanced, so the analysis is designed to detect broad sign and magnitude rather than precise subgroup or factor-model effects.

5. Current Empirical Progress

The empirical analysis was initiated before the formal proposal examination and is therefore substantially advanced. The results below document current research progress and remain subject to revision in response to advisor and committee feedback before the final degree defense.

5.1 Core Indonesia-Focused Estimate

The current 2023 base estimate is $66.83 billion of unbooked GTV against $4.18 billion of booked platform revenue, producing a 16.0× ecosystem ratio. GoTo's much higher platform-level ratio reflects both its marketplace economics and its revenue-recognition convention, so cross-platform levels are interpreted cautiously rather than as directly comparable measures of hidden economic activity.

5.2 Persistence of the Transaction-Revenue Gap

Table 5 reports a within-company GoTo series from primary Group disclosures. The 2020-2021 rows use GoTo's pro forma performance highlights as if Tokopedia had been consolidated from 1 January 2020; 2022-2023 use reported Group figures. Because GoTo also operated outside Indonesia, this series is Group-level rather than country-isolated.

Table 5. GoTo Group GTV, net revenue, and unbooked share, 2020-2023

Year	GTV	Net revenue	Unbooked GTV	Share unbooked
2020	$22.64B	$0.33B	$22.31B	98.5%
2021	$32.26B	$0.36B	$31.90B	98.9%
2022	$41.30B	$0.76B	$40.54B	98.1%
2023	$39.81B	$0.97B	$38.84B	97.6%

GoTo's unbooked share remains between 97.6% and 98.9% throughout the series. It peaks in 2021 and declines to 97.6% in 2023 as net revenue grows faster than GTV. The result is therefore persistence of a large transaction-revenue gap, not evidence that the gap widened monotonically.

5.3 Malaysia Benchmark and ASEAN-5 Scenario

The direct benchmark is Malaysia's reported LVG collection: RM476 million in 2024, or about US$104 million at the 2024 period-average exchange rate. Pre-existing Sales Tax on

Digital Services (SToDS) receipts were RM1.62 billion, or about US$354 million, in the same year. The ringgit amounts remain the source-of-record values.

The comparative benchmark uses Malaysia's SToDS plus LVG collections and Vietnam's foreign-supplier e-portal collections. Malaysia rises from RM1.15 billion in 2023 to RM2.096 billion in 2024; Vietnam rises from VND6.896 trillion to VND8.687 trillion (VietnamPlus 2023; General Department of Taxation Vietnam 2024). Each annual observation is converted using the corresponding period-average official exchange rate.

Under that convention, Malaysia rises from about US$252.1 million to US$458.0 million (+US$205.9 million), while Vietnam rises from about US$289.9 million to US$359.5 million (+US$69.6 million). The difference between those changes is about US$136.3 million. The local-currency source amounts remain the source of record, with U.S.-dollar values used only for comparability.

The comparative difference is larger because Malaysia's 2024 total includes both the new LVG stream and growth in the pre-existing SToDS stream. With only two countries and two years, the data cannot identify how much of that relative increase was caused by the LVG expansion itself.

Limitations. The comparison has two countries, two years, and no pre-trend test. More importantly, the tax bases differ: Malaysia's LVG regime taxes imported physical goods below RM500, while Vietnam's portal collects VAT and corporate income tax from nonresident digital suppliers. Vietnam is therefore a continuity benchmark, not a matched causal control. A broader comparator set, placebo timing, or a goods-import control series would strengthen future identification.

Table 6. Malaysia's 2024 descriptive collection benchmark

Series	2023	2024	Change
Malaysia (SToDS + LVG)	$252M	$458M	+$206M
Vietnam (foreign-supplier e-portal)	$290M	$359M	+$70M
Difference between changes	—	—	$136M
Malaysia LVG (RM476M source)	—	$104M	—

The US$104 million LVG collection and the US$136 million before-and-after difference are different descriptive quantities. The two-country, two-year comparison is not a causal design.

The evidentiary hierarchy is therefore straightforward. Indonesia contributes the 2023 three-platform base: $71.0 billion of GTV, $4.18 billion of platform revenue, and a $66.8 billion wedge. The four additional countries use 2024 economy-wide digital GMV because comparable three-platform data are unavailable. The resulting ASEAN-5 total of about $202 billion is a mixed-scope, mixed-year scenario anchored on the Indonesia estimate, not a second independently measured result.

Table 7. ASEAN-5 order-of-magnitude scenario

Country	Digital GMV	Implied platform revenue	Calibrated unbooked GTV	Ratio
Indonesia	$71.0B	$4.18B	$66.8B	16.0×
Vietnam	$36.0B	$2.12B	$33.9B	16.0×
Philippines	$31.0B	$1.82B	$29.2B	16.0×
Thailand	$46.0B	$2.70B	$43.3B	16.0×
Malaysia	$31.0B	$1.82B	$29.2B	16.0×
ASEAN-5	$215B	$12.64B	$202B	16.0×

The four non-Indonesian rows inherit the same 5.88% retained-share assumption. The approximately $202B total is therefore a mixed-scope, mixed-year calibration anchored on the Indonesia-focused base, not a uniformly measured regional panel.

5.4 Investor-Response Results

The event study reports eight primary Pearson specifications, and none is statistically significant. In the full sample, the level correlation is -0.138 ($p = 0.356$, $N = 47$) and the adjacent-quarter change correlation is 0.119 ($p = 0.447$, $N = 43$). Excluding the 26 strictly contaminated quarters leaves a level correlation of -0.328 ($p = 0.147$, $N = 21$) and a change correlation of 0.211 ($p = 0.401$, $N = 18$). The 2024Q1+ post-break specifications are also non-significant.

Table 8. Primary event-study correlations

Specification	N	Pearson r	p-value
Level, all observations	47	-0.138	0.356
QoQ change, all observations	43	0.119	0.447
Level, excluding strict-contaminated quarters	21	-0.328	0.147
QoQ change, excluding strict-contaminated quarters	18	0.211	0.401
Level, post-break period (2024Q1+)	27	-0.081	0.687

Specification	N	Pearson r	p-value
QoQ change, post-break period	27	-0.127	0.527

The robustness checks do not overturn the primary null. Spearman correlations are non-significant across the primary specifications, and the isolated nominal significance that appears in one very small alternative subset is not reproduced across rank-correlation or QoQ specifications. The current evidence therefore does not support a robust short-window pricing relation.

6. Interpretation, Limitations, and Expected Contribution

6.1 Interpretation Boundaries

The wedge is GTV not booked as platform revenue. It can include participant receipts, cost of goods, vehicle or labor costs, taxes already paid, refunds, and other pass-through amounts. It should therefore not be interpreted as "untaxed value creation." Estimating profit, value-added, or tax due would require participant-level cost and filing data that are not available in these disclosures.

The estimates cover selected transaction platforms and omit other forms of digital labor, creator income, direct social-commerce transactions, and activity on smaller platforms. I do not add a separate creator-economy range because no source of comparable provenance has been identified. The result is therefore not a census of platform income, but it avoids expanding the calculation with weakly sourced inputs.

The four non-Indonesian inputs in Table 7 are 2024 economy-wide digital GMV from Google, Temasek and Bain (2024): $31 billion for Malaysia, $36 billion for Vietnam, $31 billion for the Philippines, and $46 billion for Thailand. They differ in both year and scope from the 2023 three-platform Indonesia-focused base, which itself includes a Group-level GoTo input. Table 7 should therefore be interpreted as a transparent calibration scenario rather than a uniformly measured regional panel.

6.2 Policy and Academic Contribution

The evidence supports a clear distinction between information collection and tax liability. OECD platform-reporting rules and the European Union's DAC7 regime show that platforms can identify sellers, collect transaction information, and report income-relevant data to tax authorities (OECD 2019, 2020a; European Union 2021). Such reporting creates an auditable record; it does not imply that every gross payout should be taxed or withheld at the same rate.

For participants above a clearly defined threshold, the strongest implication is platform-level information reporting. Withholding may be appropriate when the tax base is simple and deductible costs can be handled, but it is not automatically superior across platform types. Because platforms already compute payouts and maintain identity and transaction records, they can lower reporting costs; whether they should also remit tax depends on local law, participant expenses, and administrative capacity (Kleven, Kreiner and Saez 2016; Slemrod 2019). The policy sequence is therefore: establish reliable reporting, distinguish participant types, and then evaluate withholding.

The ecosystem ratio measures the scale of the administrative information problem. It does not determine the appropriate tax threshold, rate, deduction rule, or social-protection design; those parameters require participant-level microdata and policy testing.

For the proposal stage, the central contribution is therefore a measurement and information-reporting argument: large platform-mediated transaction flows can be digitally recorded without being equivalent to the revenue base visible in corporate accounts. The thesis asks whether the platform's existing information infrastructure can be used to improve participant-level administrative visibility before questions of liability, withholding, deductions, and thresholds are decided.

7. Remaining Work Before Final Degree Defense

The core empirical analysis is substantially complete. The proposal examination serves as a formal review of design, scope, and interpretation before finalization. Remaining work is to incorporate committee feedback, complete any requested robustness or provenance checks, refine the measurement boundaries, and prepare the final manuscript for the degree oral examination.

7.1 Work Plan

September: proposal oral and committee feedback. October-November: requested robustness/provenance checks, advisor review, and manuscript finalization. December 2026 (or another advisor-approved date): final degree oral examination and thesis submission.

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